Chu Kai Le

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Education

National University of Singapore (NUS) Aug 2026 – May 2029
B.Eng. Mechanical Engineering, Second Major in Computing, Minor in Mathematics *Singapore*

- **Awards:** CDE Engineering Scholars Programme (Full-ride, non-bonded Scholarship)
- **Relevant Coursework:** CS2040DE Data Structures and Algorithms

NUS High School of Mathematics and Science Jan 2018 – Nov 2023
NUS High Diploma (High Distinction) *Singapore*

- **Major CAP:** Mathematics: 4.9/5.0, Chemistry: 4.9/5.0, Physics: 4.6/5.0

Experience

Genk Capital Feb 2026 – May 2026
Quantitative Research Intern *Singapore*

- Developed and debugged Python data-processing pipelines for high-frequency financial data, handling daylight-saving transitions, trading halts, early closes, and missing observations to improve dataset reliability.
- Diagnosed batch simulation failures in an open-source backtesting engine and resolved memory-related issues through explicit garbage collection between runs.
- Applied statistical hypothesis testing, Monte Carlo simulations, and block bootstrapping to analyze time-series behaviour and assess the significance of observed patterns.
- Developed predictive signals using **106 days** of high-frequency data and conducted survival analysis on **605K order-book updates**, evaluating model behaviour across market regimes and parameter settings.

Projects/Competitions

Autonomous Robot – NUS EG1311 | *Arduino, Fusion 360*

- Prototyped and Designed the chassis and catapult mechanism in Fusion 360 for a team-built Arduino-controlled autonomous robot.
- Contributed to ultrasonic sensor and DC motor integration, including motor-driver wiring and Arduino code for sensor-based feedback control.
- Tested and debugged the robot's mechanical and electronic subsystems, troubleshooting intermittent ultrasonic sensor failures during obstacle detection.

Multi-Exchange Market Data Infrastructure | *Python, Linux*

- Built and operated multi-exchange market-data collectors with compressed storage, data-quality validation, and automated archival to Cloudflare R2.
- Developed monitoring and recovery workflows for stale feeds, frozen order books, and collector failures, using automated audits to detect data-integrity issues.

IMC Prosperity 04 (2026) | *Python*

- Placed **123rd / 18,803** globally in university algorithmic trading competition; led strategy for a 5-person team of IMO/IOI/IChO gold medalists.

Technical Skills/Awards

Programming: Python (Pandas, NumPy, Numba), SQL, Bash
Engineering: Arduino, Fusion 360, CAD modelling, sensor and motor integration
Systems & Tools: Linux, Git, SSH, real-time data pipelines
Awards: Singapore Armed Forces National Serviceman of the Year (2026)
Interests: Valorant (Peak Top 0.8%), Trackmania (Peak Top 1 SG, Top 1% Asia), Bouldering (V7), Motorcycling